

MOHAMMED GHALIB

Software & AI Engineer • End-to-End Product Delivery • RAG, Agentic Systems, Django / React

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PROFESSIONAL SUMMARY

Software and AI engineer who builds and ships complete products, not just features. I write the Django backends, the React web and React Native mobile apps on top of them, and the auth, payments, and background jobs underneath, then containerise, deploy, and keep the whole thing running in production. I take a product from the first architecture decision through to a live, maintained release, working directly with the client or team it is for.

I build the AI too, and I build it to be reliable. When a product needs it, I add retrieval and agentic systems that plan their own lookups, ground every answer in real sources, verify their own output, and refuse to guess when they aren't sure. That is the core of Sheria AI, a Kenyan legal research platform, and Ummly, an Islamic knowledge platform, and I have taken the same approach into client work across healthcare, education, and IoT.

I currently lead an AI legal research product end to end at Datawise Africa. Before that I carried client projects at Digital Apex from the first requirements call to live handover, and ran the internal platform and product AI at Bee MultiScent.

PROFESSIONAL EXPERIENCE

Software & AI Engineer, Datawise Africa

Feb 2026 – Present • Nairobi / Remote

Kenyan software and AI company. I lead Sheria AI end to end and build in-house systems with the team.

- **Lead Sheria AI** (sheria.africa), a legal research platform for Kenyan law. Handed to me as a project to lead, and delivered as a working platform now in production. It takes a question in plain language and answers it by citing, quoting, and explaining the actual provision from the Constitution, statutes, and decided cases, so advocates, in-house teams, and law students find the authority in minutes instead of an afternoon of reading.
- **Rebuilt the core** from a single retrieve-then-answer call into an **agentic research pipeline on LangGraph**: each question is triaged, then researched by a model that plans its own lookups against case-law and Constitution tools, and each answer is drafted, verified, and repaired before the user sees it, with a coverage check so no part of a multi-part question is dropped.
- **Kept the results domain-specific and grounded**. Chunked the case index so search reads the court's actual reasoning rather than the header, tuned the relevance thresholds that keep unrelated matters out of answers, added exact-name case lookup, and enforced cited sources with a firm no-fabrication rule and a plain refusal when nothing relevant exists.
- **Built the evaluation harnesses** the team ships on: a triage scorecard, and a blind side-by-side comparison of candidate answer models, so prompt and model changes go out on evidence rather than opinion. The judgment lives in editable prompt files, so behaviour is tuned in prose rather than code.
- **Own the frontend**, built with React Router (server-side rendering), TypeScript, Redux Toolkit, and TanStack Query: a streaming chat UI that renders answers as they generate, with ranked source cards.
- **Deliver in-house projects across the academic and health sectors** as team work: shared repositories on Git and GitHub, branching and pull-request review, CI, a quality gate, test coverage, and versioned releases.

Software Engineer, Digital Apex

Dec 2024 – Jan 2026 • Remote

Kenyan technology company building its own products and delivering projects for clients. I took on incoming client work and maintained the company's internal systems.

- **Carried client projects end to end**, from the first requirements conversation through architecture, build, deployment, and handover: chatbots, retrieval-based (RAG) assistants, internal business systems, and operational dashboards, for organisations including healthcare and education providers.
- **Built and maintained the company platform** (digitalappex.co.ke), a Django and DRF backend with a React frontend covering products, orders, quotes, payments, warranties, messaging, projects, and a chatbot. It stayed live throughout: feature work, maintenance, and releases.
- **Built the foundations the products run on**: JWT auth, Paystack payments, role-based access, and background jobs on Celery and Redis, containerised with Docker and deployed to a DigitalOcean server behind nginx-proxy with automated SSL and CI on GitHub Actions.
- **Set engineering standards across the team's work**: architecture reviews, documentation, and a repeatable release process, so a project could be handed over and maintained by someone other than the person who wrote it.

Software Engineer, Bee MultiScent Ltd

Feb 2023 – Dec 2024 • Hybrid

Scent-technology company. I ran the internal platform, the AI diffuser work, and the website.

- **Built the company portal**: a Django and DRF backend with a React frontend and real-time dashboards used across operations, HR, and inventory, with role-based access, task assignment, document management, and WebSocket notifications, plus a companion React Native mobile app on the same API.

- **Developed the AI for a smart scent diffuser**, automating how and when the company's IoT diffuser released scent.
- **Built and maintained the company website**, and owned ongoing development across the company's systems. Earned an official recommendation letter for performance and teamwork (available on request).

SELECTED PROJECTS

Independent and contract work delivered alongside the roles above.

- **Ummly (formerly Islamify), Islamic knowledge platform** (ummly.co.ke). Built the whole product solo across **backend, web, and mobile**, around a **Pinecone RAG** assistant grounded in the Qur'an and Hadith: reranking, a confidence floor that returns a low-confidence message instead of guessing, intent classification, query rewriting, bilingual EN/AR answers, and per-excerpt citations. LLaMA via Groq with a Google Gemini fallback. Containerised and deployed through GitHub Actions to a DigitalOcean server, with email and Paystack payments, behind a React Router (SSR) web app on Vercel.
- **Ummly mobile app** (Expo / React Native, TypeScript). Built end to end against my own DRF API: TanStack Query for server state, caching, infinite-scroll feeds, and optimistic updates, and JWT in the device secure store with a single-flight refresh that replays the failed request on a 401 so sessions never drop mid-action.
- **Collaborative Approach, a referral and tour-booking SaaS** (collaborativeapproach.africa). React and DRF, with M-Pesa STK Push, PayPal, and Stripe payments, referral tracking with automated commission calculation, Celery background tasks, and role-based portals (Admin / Agent / Customer).
- **Sheikh Siraj Rahman College, a school management system** (sheikhsirajrahman.ac.ke). Multi-portal (Admin / Teacher / Student / Parent) on React, Django, and PostgreSQL, with online admissions, fee payments, results publishing, and SMS and email automation, on a CI/CD pipeline.

CORE SKILLS

Product Ownership: End-to-end delivery from requirements to live release, architecture and technical decisions, direct client and stakeholder communication, scoping, estimation, documentation, and handover

Engineering Leadership: Setting standards a team can build on, code review, release management, keeping production systems healthy, and mentoring through review

Applied AI Judgment: Turning a business problem into a grounded, evaluated AI system rather than a demo: knowing where a model will fail, and proving it before users do

Ways of Working: Agile/Scrum, Git and GitHub, branching and pull-request review, Jira, Slack, remote and hybrid teams

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript/JavaScript, Django, Django REST Framework, FastAPI, Celery, React, React Router (SSR), React Native (Expo), Tailwind CSS, Redux Toolkit, TanStack Query

AI Engineering: Agentic pipelines on LangGraph (plan, research, verify, repair), tool-calling, RAG design, LangChain, vector databases (Pinecone, ChromaDB), chunked embeddings, reranking (bge-reranker), similarity thresholds, grounding & citation, hallucination control, prompt engineering, evaluation harnesses

LLMs: OpenAI, LLaMA (via Groq), Google Gemini, DeepSeek, reasoning models (gpt-oss), multi-model routing and fallback

APIs & Integrations: REST APIs, JWT (RS256 / SimpleJWT), WebSockets/Channels, Paystack, M-Pesa STK Push, PayPal, Stripe, ZeptoMail, SMS and email gateways

Data & Infra: PostgreSQL, Redis, SQLite, vector stores, Docker & docker-compose, Gunicorn, Nginx, DigitalOcean, Vercel, Render, Supabase

DevOps & Quality: GitHub Actions CI/CD, SonarQube quality gates, PyTest, Vitest, versioned releases, staging and production environments

EDUCATION

Bachelor of Business Information Technology, Mount Kenya University

Information Technology, Software Development, and Artificial Intelligence.

CERTIFICATIONS

- Backend Engineer Certification
- AI Engineer Certification

REFERENCES

Available upon request.